CH32M030 Evaluation Board Reference

Version: V1.1

https://wch-ic.com

1. Overview

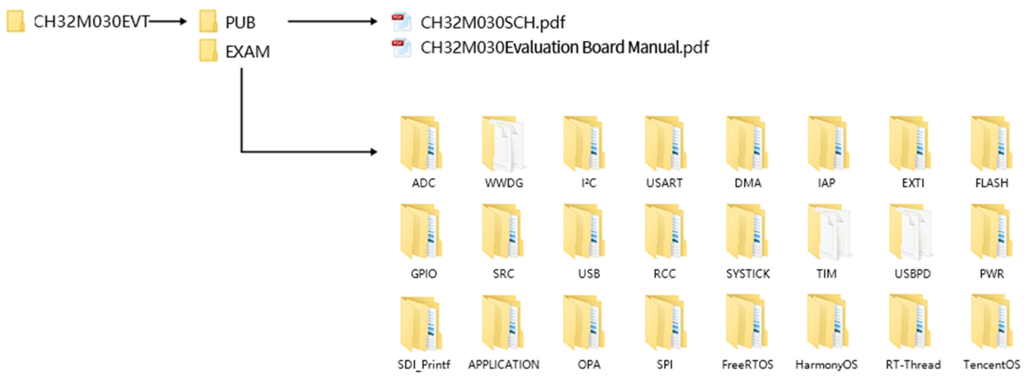
This evaluation board is applied to the development of CH32M030 series chips, the IDE uses MounRiver compiler, supported only by MRS1.92 and above, and independent WCH-Link can be selected for simulation and download, and application reference examples and demos related to chip resources are provided.

Please refer to the CH32M030SCH.pdf document for the schematic of the evaluation board.

The debug interface of CH32M030 series chips supports free configuration; 1-wire debugging or 2-wire debugging is optional. Debugging interface pins PA3 (SWIO), PA2 (SWCLK 2-wire debugging optional)

2. Software Development

2.1 EVT package directory structure



Description.

PUB folder: Provides evaluation board manuals, schematics for evaluation versions, and chip support package library files.

EXAM folder: Provides software development drivers and corresponding examples for the CH32M030 controller, categorized by peripherals. Each type of peripheral folder contains one or more functional application routines folders.

2.2 IDE Use-MounRiver

Download MounRiver_Studio, double click to install it, and you can use it after installation. (MounRiver_Studio instructions are available at the path: MounRiver\MounRiver_Studio\ MounRiver_Help.pdf and MounRiver_ToolbarHelp.pdf)

2.2.1 Open Project

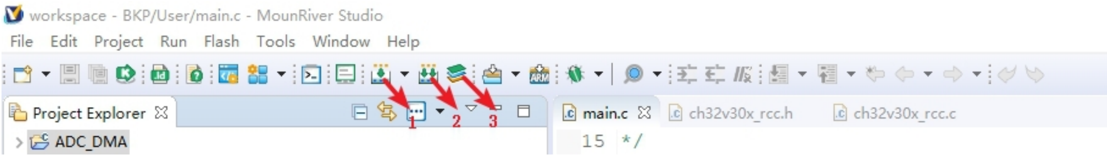
➢ Open project:

1) Double-click project file directly with the suffix name .wvproj under the corresponding project path.
2) Click File in MounRiver IDE, click Load Project, select the .project file under the corresponding path, and

click Confirm to apply it.

2.2.2 Compilation

MounRiver contains 3 compilation options, as shown in the following figure.



Compile option 1 is Incremental Build, which compiles the modified parts of the selected project.

Compile option 2 is ReBuild, which performs a global compilation of the selected project.

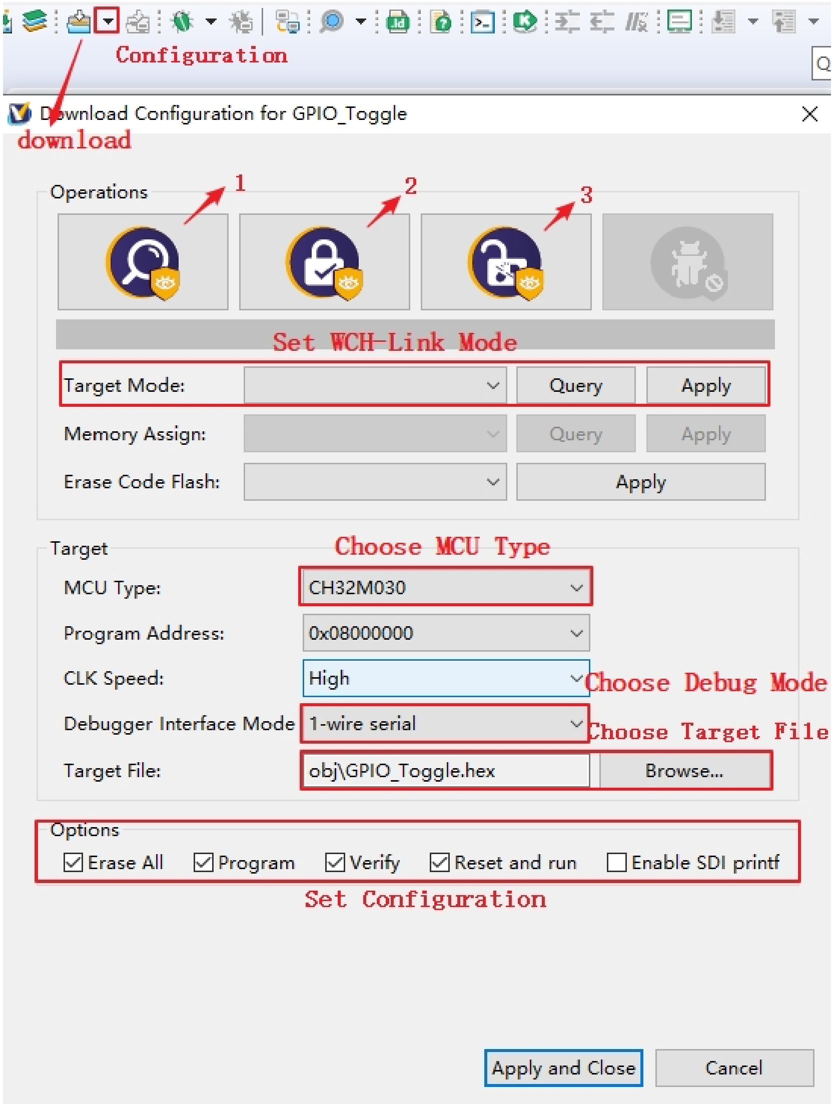
Compile option 3 is All Build, which performs global compilation for all projects.

2.2.3 Download/Simulation

➢ Download

1) Debugger download

Connect to the hardware via WCH-Link (see WCH-Link instructions for details, path: MounRiver\MounRiver_Studio\ WCH-Link instructions.pdf), click the Download button on the IDE, and select Download in the pop-up interface, as shown in the figure below.



1 for querying the chip read protection status.
2 for setting the chip read protection and re-powering the configuration to take effect.
3 for lifting the chip read protection and re-powering the configuration to take effect.

- Simulation

1) Toolbar description

Click Debug button in the menu bar to enter the download, see the image below, the download toolbar.

Detailed functions are as follows.

(1) Restart: After restart, the program returns to the very beginning.
(2) Continue: Click to continue debugging.
(3) Terminate: Click to exit debugging.
(4) Single-step jump-in: Each time you tap a key, the program runs one step and encounters a function to enter and execute.
(5) Single-step skip: jump out of the function and prepare the next statement.
(6) Single-step return: return the function you jumped into

Instruction set single-step mode: click to enter instruction set debugging (need to use with 4, 5 and 6 functions).

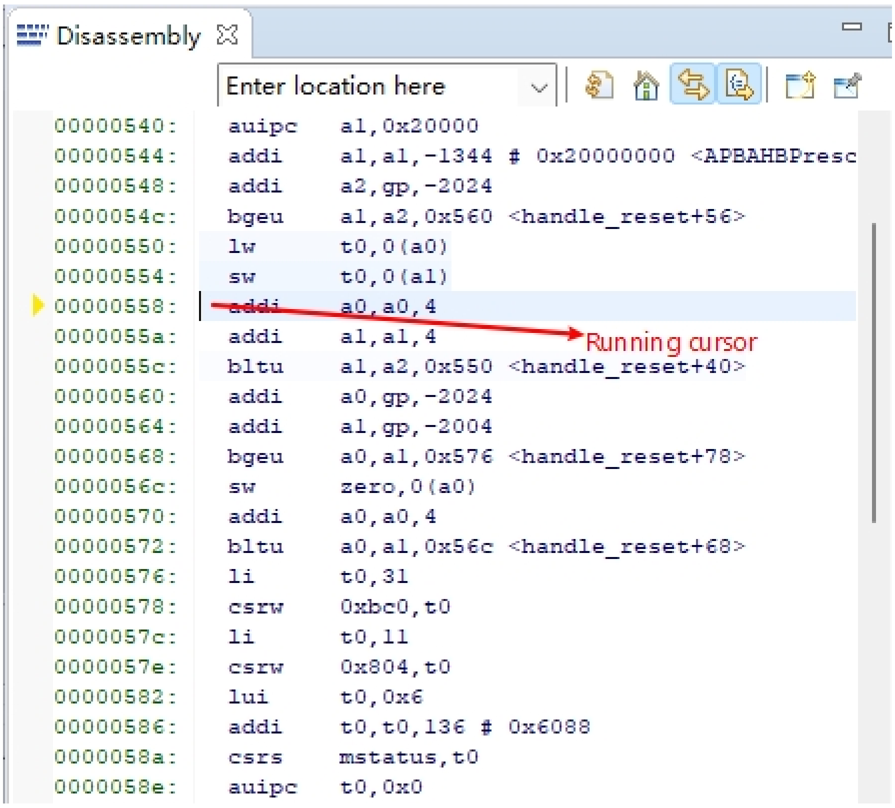
2) Set breakpoints

Double-click on the left side of the code to set a breakpoint, double click again to cancel the breakpoint, set the breakpoint as shown in the following figure;

3) Interface display

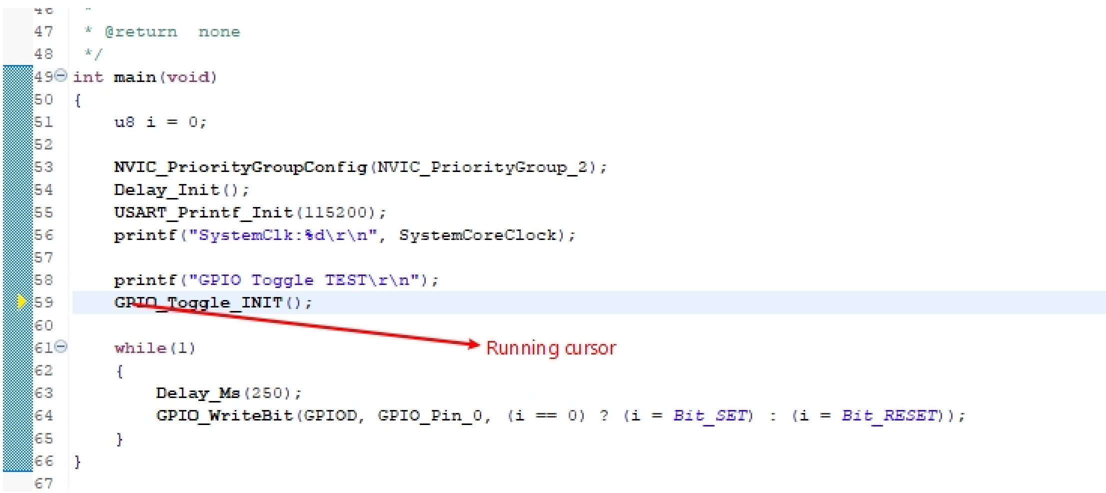
(1) Instruction set interface

Click on the instruction set single-step debugging can enter the instruction debugging, to single-step jump in for example, click once to run once, the running cursor will move to view the program running, the instruction set interface is shown as follows.



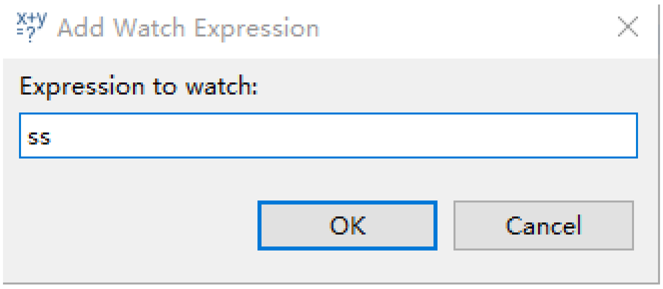
(2) Program running interface

It can be used with instruction set single-step debugging, still take single-step jumping in as an example, click once to run once, the running cursor will move to view the program running, the program running interface is shown as follows.

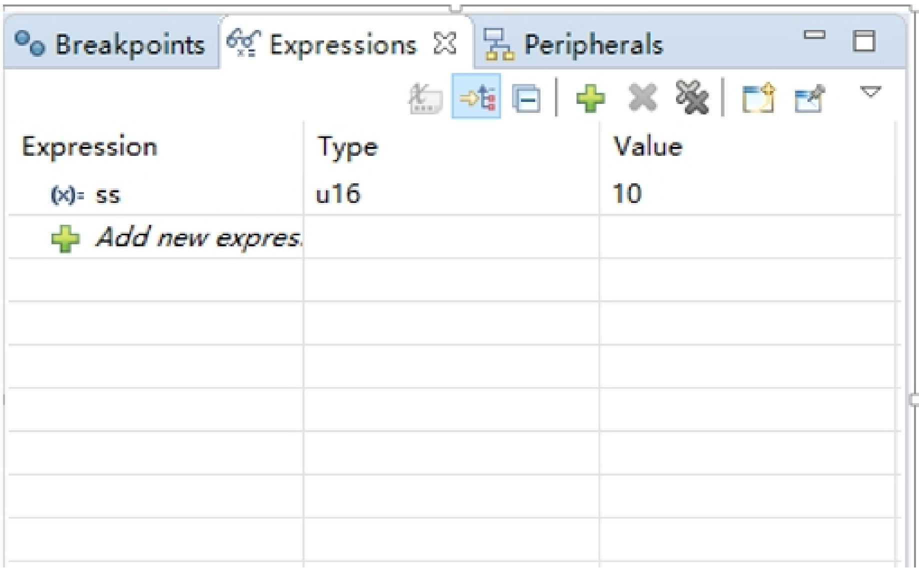


4) Variables

Hover over the variable in the source code to display the details, or select the variable and right-click add watch expression

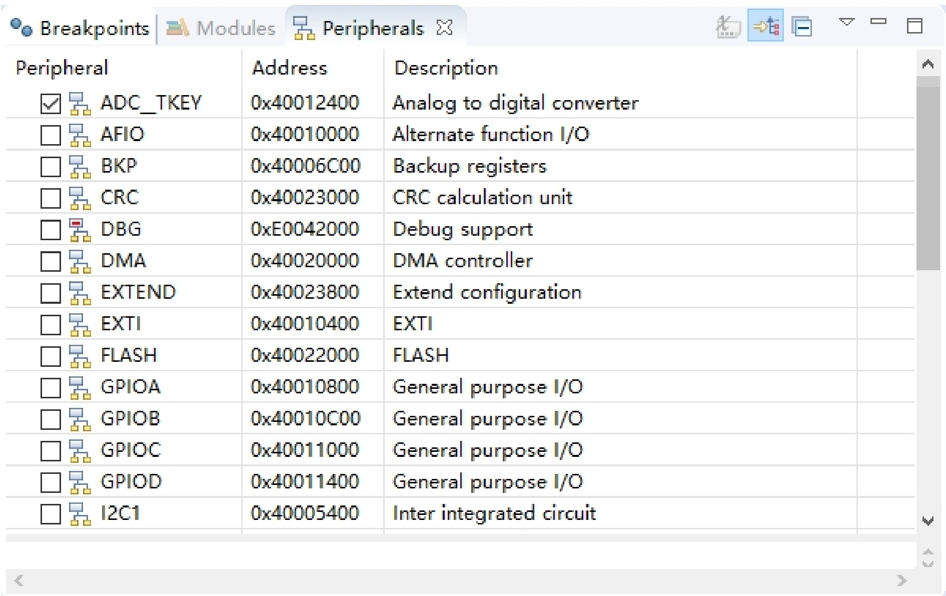


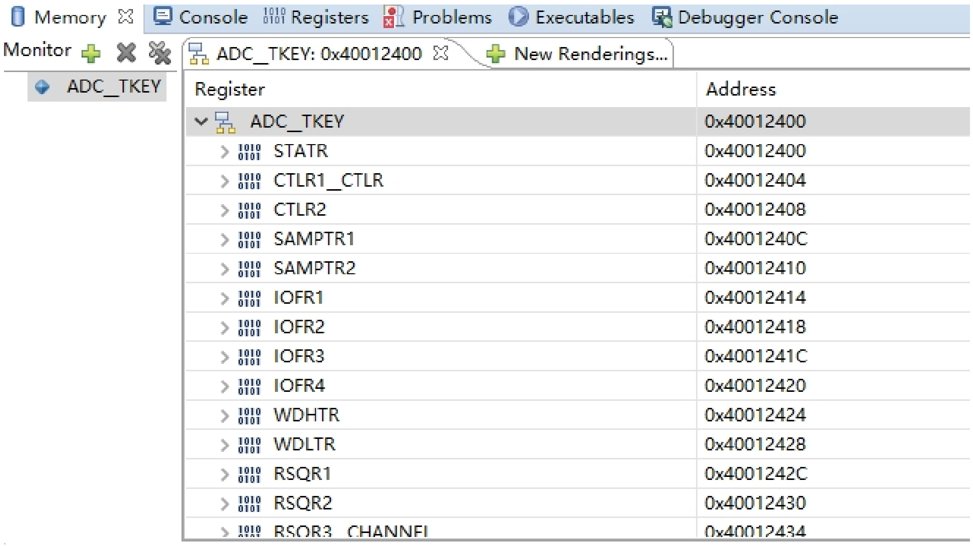
Fill in the variable name, or just click OK to add the variable you just selected to the pop-up.



5) Peripheral registers

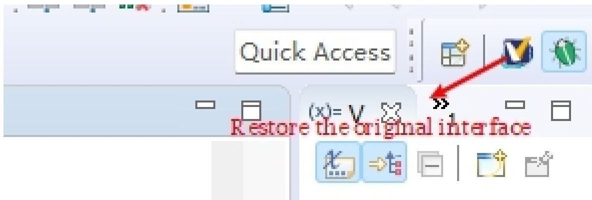
In the lower left corner of IDE interface Peripherals interface shows a list of peripherals, tick the peripherals will display its specific register name, address, value in the Memory window.





Note:

(1) When debugging, click the icon in the upper right corner to enter the original interface.

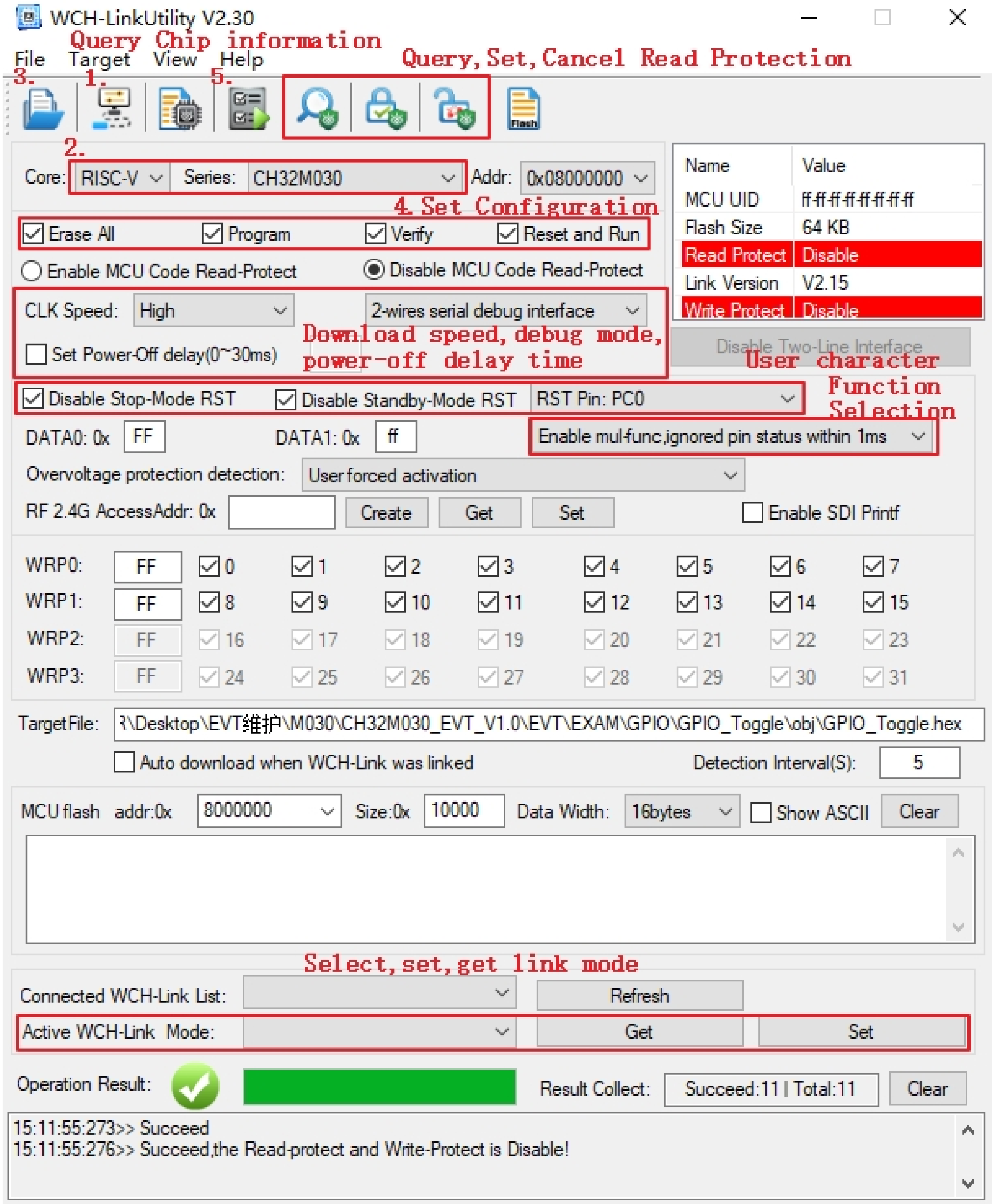


(2) For documentation to access the compiler, click F1 to access the help documentation for detailed instructions.

3. WCH-LinkUtility.exe Download

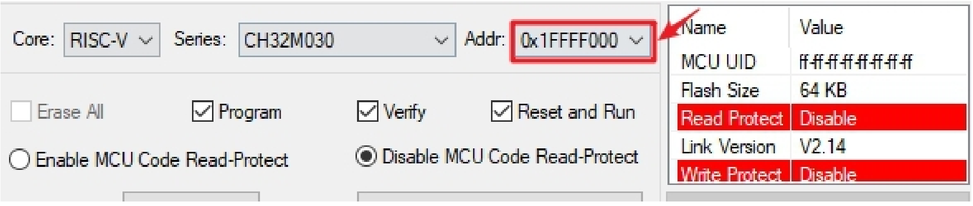
The download process for the chip using the WCH-LinkUtility tool is:

1) Connect WCH-Link
2) Select chip information
3) Add firmware
4) If the chip is read protected, you need to release the chip read protection.
5) Execute



The CH32M030 chip supports a 512-byte DATA FLASH area, and the data will not be lost when power is off. The address range is 0x1FFFF000~0x1FFFF1FF; Only the LinkUtility tool can be used to operate the DATA FLASH area. Select the download address as 0x1FFFF000 (as shown in the figure), and select the custom bin file to be stored. After downloading, the values in the bin file can be stored in the DATA FLASH area. Follow steps 3, 4, and

5 to add firmware, select configuration, and perform download.



Tips: Note that the custom bin file size must be 512 bytes, and the unused area is written as 1 by default.

4. Related Links

WCH official website: https://www.wch-ic.com/

WCH-LINK instructions for use: https://www.wch-ic.com/products/WCH-Link.html